

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 clocks, debug and I/O allocation



File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 I/O and analog supplies



File: mcu_supply_power.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

Sheet: / File: ereader_rev1.kicad_sch			
Title: RA8P1 e-reader - schematic index			
Size: A3	Date: 2026-09-05		Rev: 0.2 DRAFT
KiCad E.D.A. 10.0.5		Id: 1/4	

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DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

U1A
R7KA8P1KFLCAC#UC0

Clock / Reset / Debug

P201/_MD P212/E_XTAL K17
P208/T_DI P213/X_TAL K16
P209/T_D0 P214/X_COUT L16
P210/T_M5/S_WDIO P215/X_CIN/EX_CIN L17
P211/T_CK/S_WCLK
RES

U1B
R7KA8P1KFLCAC#UC0

GPIO P0 / P1

P000 P100 U6
P001 P101 R5
P002 P102 P5
P003 P103 R4
P004 P104 M6
P005 P105 N7
P006 P106 N6
P007 P107 N5
P008 P108 D8
P009 P109 D7
P010 P110 C8
P011 P111 E8
P012 P112 A4
P013 P113 A2
P014 P114 B3
P015 P115 A3

U1C
R7KA8P1KFLCAC#UC0

GPIO P2 / P3

P200 P300 B5
P206 P301 C4
P207 P302 A5
P303 B6
P304 D10
P305 C10
P306 D11
P307 C11
P308 C9
P309 A12
P310 E10
P311 B12
P312 C13

U1D
R7KA8P1KFLCAC#UC0

GPIO P4 / P5

P400 P500 U7
P401 P501 R8
P402 P502 T7
P403 P503 H2
P404 P504 H1
P405 P505 H3
P406 P506 J1
P407 P507 J2
P408 P508 J3
P409 P509 J4
P410 P510 K3
P411 P511 U15
P412 P512 P13
P413 P513 R13
P414 P514 R14
P415 P515 P15

U1E
R7KA8P1KFLCAC#UC0

GPIO P6 / P7

P600 P700 F12
P601 P701 F15
P602 P702 F13
P603 P703 G14
P604 P704 G13
P605 P705 F17
P606 P706 E14
P607 P707 F16
P608 P708 P16
P609 P709 M13
P610 P710 M15
P611 P711 N15
P612 P712 N13
P613 P713 N16
P614 P714 N14
P615 P715 T15

U1F
R7KA8P1KFLCAC#UC0

GPIO P8 / P9

P800 P902 E9
P801 P903 D9
P802 P904 A16
P803 P905 A14
P804 P906 A13
P805 P907 A15
P806 P908 B13
P807 P909 B14
P808 P910 F11
P809 P911 C12
P810 P912 D12
P811 P913 E12
P812 P914 E5
P813 P915 A6

U1G
R7KA8P1KFLCAC#UC0

GPIO PA / PB

PA00 PB00 E16
PA01 PB01 D17
PA02 PB02 E13
PA03 PB03 D16
PA04 PB04 D13
PA05 PB05 D15
PA06 PB06 E14
PA07 PB07 D14
PA08
PA09
PA10
PA11
PA12
PA13
PA14
PA15

U1H
R7KA8P1KFLCAC#UC0

GPIO PC / PD

PC00 PD00 K4
PC01 PD01 B16
PC02 PD02 B17
PC03 PD03 C15
PC04 PD04 C14
PC05 PD05 C16
PC06 PD06 C17
PC07 PD07 E15
PC08
PC09
PC10
PC11
PC12
PC13
PC14
PC15

U1M
R7KA8P1KFLCAC#UC0

USB / MIPI

AVCC_USBH_S AVCC_MIPI T4
P814/U_SB_DP MIPI_CL_N U2
P815/U_SB_D_M MIPI_CL_P T2
USBH_S_DM MIPI_DLO_N U1
USBH_S_DP MIPI_DLO_P T1
USBH_S_RRE_F MIPI_DL1_N U3
VCC_U_SB MIPI_DL1_P T3
VCC_U_SBHS VCC18_MIPI R2
VSS_U_SB VSS_MIPI R3
VSS1_USBH_S
VSS2_USBH_S

RA8P1 e--reader

Sheet: /RA8P1 clocks, debug and I/O allocation/
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